

Doctor Apple — Hack4Health 2026 Technical Track Submission

1. Team Information

- **Team name:** Doctor Apple
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2. Executive Summary (200 words maximum)

Doctor Apple is an AI-assisted clinic pre-registration system connecting patients, clinic staff, and third-party administrators (TPAs). Patients sign in with the email already attached to their synthetic patient record, choose a booked or walk-in visit, upload a referral chit or insurance voucher from a phone or computer, and review a questionnaire prefilled from existing general or occupational-health records. Agnes AI extracts document fields, while deterministic rules—not the language model—match identity, policy, package eligibility, covered services, and claim limits.

Uncertain, inconsistent, or unsupported cases are routed to a human with an explicit reason. Immutable identity fields remain protected, while permitted questionnaire answers can be corrected and saved back to the patient record. Clinic staff must verify physical identity and the insurance e-card in person before approval.

After services are recorded, eligible claims can be automatically approved or rejected with reasons and reflected in the TPA and clinic views; exceptions remain in manual review. The prototype provides role-based portals, mobile-responsive upload and forms, audit events, masked identifiers, MongoDB persistence with an offline test store, and an initial Microsoft Copilot Studio agent. For the Copilot integration, we have successfully tested patient profile retrieval and appointment history tracking, with a documented path for migrating more complex workflows. (189 words)

3. Problem Understanding

Clinic pre-registration currently requires patients and counter staff to repeatedly transcribe identity, referral, policy, appointment, and health-questionnaire information across disconnected steps. Walk-ins add queue pressure; booked patients can still arrive with incomplete or mismatched documents. Staff must interpret different insurers' package codes and then communicate claim status to the TPA and clinic.

The root causes are fragmented records, manual document reading, policy variation, and absent end-to-end status synchronization. These cause longer queues, duplicate entry, privacy exposure, incorrect billing, and avoidable manual follow-up. In health screening, mistakes can also hide allergies or send a patient to the wrong package, making reliable exception handling essential.

4. Proposed AI Solution

The patient authenticates, selects walk-in or prebooked care, chooses general or occupational screening, and supplies a local PDF/photo or chooses self-pay. Agnes AI performs schema-constrained document extraction. Doctor Apple's deterministic services then confirm that the document belongs to the signed-in patient, resolve the insurer/package, compare requested services with coverage, prefill the appropriate Parkway Shenton questionnaire, and explain the outcome.

Patients may edit and save questionnaire answers except protected name, identifier, date of birth, and gender. Staff see the registration and must complete physical identity/e-card verification. Following service delivery, the rules engine sends an eligible claim into the simulated TPA workflow and returns approved, rejected, or manual review, together with reasons, to both TPA and clinic views. Self-pay is labelled **No insurance / self-pay** and bypasses the TPA.

(Refer to our visual roadmap in the Appendix: **doctorappleuserjourneyflow_v2.jpg** for the complete, end-to-end interactive journey).

What's Hot with Doctor Apple (Core Value Proposition)

- **Dual-Engine Precision:** We decouple the AI LLM from clinical decision-making. Agnes AI handles the messy, unstructured document bóc tách (OCR), while a zero-hallucination Python rules engine executes identity matching and coverage adjudication. Financial and medical safety is 100% guaranteed.
 - **Tested Copilot Integration:** Unlike theoretical designs, we built a working Microsoft Copilot Studio agent and tested the foundation of our custom connector—successfully verifying **get patient profile** and **get appointment history** over live webhooks.
 - **Elimination of Double-Entry:** The system bridges the clinic portal and the TPA interface seamlessly, ensuring data is captured once during pre-registration and securely propagated upon staff verification.
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5. Technical Architecture

Tech Stack

- **Frontend Interface:** Built with **Streamlit** (Python 3.12) and responsive **HTML5/TailwindCSS** to provide sleek, dedicated workspaces for three distinct roles: Patient, Clinic Staff, and TPA Assessor.
- **Backend API & Orchestration:** Developed using **FastAPI** to handle fast, secure web requests with standard bearer-token authorization.
- **AI Engine:** Powered by the **Agnes OpenAI-compatible multimodal API** to perform strict schema-constrained text and image bóc tách from medical chits.
- **Database & Storage:** **MongoDB Atlas** for secure production records, coupled with a robust **MemoryStore (In-memory adapter)** supporting deterministic offline testing.
- **Enterprise Integration:** **Microsoft Copilot Studio** utilizing customized OpenAPI schemas, **Power Apps Custom Connectors**, and Power Platform flow configurations.

Deployment & System Integration

- **GitHub Repository:** Host of our complete source code and automated tests:
<https://github.com/terramii/doctorapple1999>
 - **Vercel Cloud Deployment:** Our production FastAPI backend and web prototypes are live and accessible at
<https://doctorapple1999.vercel.app/>
 - **OpenAPI Schema Sync:** FastAPI automatically generates an OpenAPI 3.0.0-compliant **openapi.json** at our production URL, which was successfully imported as a Power Platform Custom Connector to establish a direct connection between Microsoft Copilot Studio on the cloud and our backend.
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6. Operational Impact

The prototype is designed to reduce repeated transcription, incomplete forms, and policy lookups. Pilot targets (to validate rather than present as measured results) are:

- At least **50% less counter data-entry time** per pre-registered patient.
- At least **30% shorter pre-registration handling time**.
- **100% of low-confidence, identity-conflict, uncovered-service, and unknown-policy cases** routed to an explained review state.

Audit events cover registration, questionnaire submission, physical verification, and TPA decision. A clinic pilot should baseline median handling time, correction rate, manual-review rate, and claim turnaround, then compare four weeks before and after rollout.

7. Feasibility

The current local prototype implements core backend logic (including lookup, upload, questionnaire pre-fill, and staff portals). However, full integration of the Copilot Studio agent with the web frontend is still in progress. In our Copilot environment, we have successfully integrated and verified two foundational agent features: **get patient profile** and **get appointment history** via our custom connector. Advanced integration stages (OCR-based chat extraction, dynamic questionnaires, and TPA synchronization) are designed but not yet fully tested on the agent due to a lack of sufficiently diverse sample data for these complex edge cases.

Future Enhancements & Areas for Improvement

- **Hospital & Staff Literature References:** To build trust among healthcare administrators, we plan to align Doctor Apple's pre-registration efficiency metrics with published clinical workflow papers (e.g., standard clinical triage protocols and administrative efficacy studies). Adding a dedicated "Staff Knowledge Portal" with searchable medical and administrative reference guidelines will empower front-desk workers.
- **Better Agent Recognition & Training:** We aim to implement advanced Natural Language Understanding (NLU) features within Copilot Studio, such as:
 - Intent clustering and synonym mapping to handle colloquial terminology, abbreviations, and multilingual patient inputs (e.g., mixing English, Vietnamese, or local terms).

- Few-shot prompt training to optimize how the agent handles low-confidence document uploads before routing them to staff.

Implementation Roadmap (8–12 weeks)

- **2 weeks** for workflow/data mapping.
- **3 weeks** for connector and identity integration.
- **2 weeks** for security and user acceptance testing.
- **2 weeks** for a limited clinic pilot.
- **1 to 3 weeks** for remediation.

Dependencies include approved hosting, Entra ID, MongoDB or an approved clinical data store, Clinic Assist/NEHR and insurer API access, data-protection review, clinic champions, and TPA rule ownership.

8. Governance & Safety

Doctor Apple is an administrative assistant, not a diagnostic tool. Physical ID/e-card verification and human exception handling remain mandatory. Deterministic code is authoritative for coverage and billing; malformed files, low extraction confidence, conflicting identity data, unknown codes, uncovered services, and missing limits cannot be silently approved.

The prototype uses synthetic data only. Production requires explicit consent and purpose limitation, least-privilege role access, encryption in transit and at rest, managed secrets, retention/deletion rules, upload malware scanning, rate limits, incident handling, access reviews, and a PDPA data-protection impact assessment. The API is designed to minimize unnecessary exposure of patient identifiers in responses and audit records. Material actions create timestamped audit events with actor, decision, and reasons. Human reviewers can resolve exceptions, and rule/model versions should be stored with every production decision.

9. Scalability

The API/rules/UI separation allows Parkway Shenton clinics to share a governed service while configuring clinic locations, packages, insurers, limits, and questionnaires as versioned data. Stateless API replicas and a managed database can scale horizontally.

Expansion across IHH should proceed by country and workflow because privacy law, identifiers, insurers, terminology, and clinical-system interfaces differ. New rules require tests and staged approval; aggregate operational metrics should avoid exposing patient data.

10. Appendix (optional; excluded from page limit)

- **Production Deployment:** Live Web App & API: <https://doctorapple1999.vercel.app/>
- **GitHub Repository:** <https://github.com/terramii/doctorapple1999>

- **User Journey Flowchart Diagram:** Included file `doctorappleuserjourneyflow_v2.jpg`
- **Copilot Studio integration guide:** `doctor-apple-agent/COPILOT_STUDIO.md`
- **Agent manifest:** `doctor-apple-agent/agents-cli-manifest.yaml`
- **Automated evidence:** `doctor-apple-agent/tests/`
- **Prototype UI:** `sample_app/doctor-apple-sage.html`
- **Judging evidence map:** `SUBMISSION_CHECKLIST.md`